SIMULATION TOOL CISCO PACKET TRACER

Experiment: Installation and Basic Network Configuration Using Cisco Packet Tracer

AIM:

To install Cisco Packet Tracer and simulate a small local area network (LAN) using end devices and a switch, assign IPv4 addresses, and test communication between the computers.

REQUIREMENTS:

A computer or laptop, an internet connection, a Cisco Networking Academy or Skills for All account, and the Cisco Packet Tracer application.

INTRODUCTION:

Cisco Packet Tracer is a network-learning and simulation application. It allows users to build a virtual network by placing devices, connecting them with cables, configuring network settings, and testing connectivity. It is useful for practising basic networking concepts without requiring physical networking equipment.

INSTALLATION PROCEDURE:

Step 1: Visit the Cisco Networking Academy or Skills for All website and sign in with an account.

Step 2: Locate the Cisco Packet Tracer download page and choose the installer compatible with the computer's operating system.

Step 3: Wait for the installer to download, then open the downloaded setup file.

Step 4: Read and accept the license agreement to continue.

Step 5: Choose the destination folder, or retain the default installation location.

Step 6: Follow the setup wizard prompts and select the required shortcuts.

Step 7: Select Install and wait for the installation process to finish.

Step 8: Select Finish, launch Packet Tracer, and sign in if prompted.

PROCEDURE FOR BASIC NETWORK SETUP:

1. Launch Cisco Packet Tracer and open a new workspace.

2. From the End Devices category, drag two PCs (PC0 and PC1) into the workspace.

3. From Network Devices, select Switches and place one switch in the workspace.

4. Choose a Copper Straight-Through cable from the Connections menu.

5. Connect PC0 to FastEthernet port 0/1 on the switch and PC1 to FastEthernet port 0/2.

6. Select PC0, open Desktop, choose IP Configuration, and enter IP address 192.168.10.10 with subnet mask 255.255.255.0.

7. Select PC1 and enter IP address 192.168.10.11 with subnet mask 255.255.255.0.

8. Wait until the link indicators show that the connections are active.

9. On PC0, open Desktop → Command Prompt and run: ping 192.168.10.11

10. Check the replies to confirm whether PC0 can communicate with PC1.

NETWORK CONFIGURATION:

Device	IPv4 Address	Subnet Mask
PC0	192.168.10.10	255.255.255.0
PC1	192.168.10.11	255.255.255.0

VERIFICATION:

The two PCs are configured in the same IP network (192.168.10.0/24).

The ping command from PC0 to PC1 is used to test connectivity. Successful replies indicate that the simulated LAN is communicating correctly.

RESULT:

Cisco Packet Tracer was installed and used to create a basic LAN with two PCs connected through a switch. IPv4 addresses were configured, and connectivity was tested using the ping command.